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MSc Data Science

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Electricity Consumption Forecasting Using Classical and Machine Learning Time Series Models

By

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Declaration Statement

I am submitting this report to partially fulfill the requirements for the Master of Science in Data Science degree at the University of Hertfordshire.

I confirm that I have carefully reviewed the university's guidelines regarding academic integrity, plagiarism, and assessment offenses. I fully understand how the university handles suspected academic misconduct and am aware that potential penalties include failing this project or the entire course.

I declare that this submission is entirely my own original work. Any ideas, data, or quotes sourced from the published or unpublished work of others have been properly cited and fully acknowledged in accordance with university regulations (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6).

Furthermore, I confirm that no human participants were involved in the research for this MSc project.

Finally, I grant permission for this report to be shared on university module websites, provided that my authorship is clearly acknowledged.

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Acknowledgement

This research stands as a major achievement in my education, proving the analytical and practical abilities I have gained through my coursework. The project allowed me to turn classroom theory into real-world practice, deeply improving both my technical skills and my ability to solve complex problems.

I want to express my genuine gratitude to my supervisor, Vito Graffagnino, for his steady encouragement, sharp feedback, and excellent guidance from start to finish. His leadership, combined with the strong academic tools and supportive environment offered by the University of Hertfordshire, were absolutely critical in bringing this research to life.

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Abstract

This research investigates the prediction of domestic energy usage by evaluating both traditional statistical frameworks and contemporary machine learning algorithms. Utilizing the UCI Individual Household Electric Power Consumption dataset, the analysis specifically targets global active power. To facilitate robust modeling, a comprehensive analytical pipeline was established. This methodology encompassed the mitigation of missing data via chronologically based interpolation, the reduction of high-resolution sensor logs to hourly aggregates, and the extraction of temporal patterns through feature engineering, specifically leveraging lagging indicators, rolling window statistics, and date-specific parameters.

An array of predictive frameworks was constructed to systematically benchmark forecasting efficacy. The evaluation suite comprised rudimentary baselines (standard and seasonal naïve approaches), a conventional SARIMA architecture, and robust machine learning methodologies, namely Ridge Regression, Random Forest, and XGBoost. The comparative analysis relied on established error metricsMean Absolute Error (MAE), Root Mean Square Error (RMSE), and symmetric Mean Absolute Percentage Error (sMAPE)applied systematically across two distinct predictive paradigms: single-step and iterative, recursive forecasting.

The empirical findings reveal a significant divergence in algorithmic performance dependent upon the forecasting framework. Although sophisticated techniques such as XGBoost exhibit superior accuracy in isolated, single-step projections, their predictive reliability degrades considerably when subjected to the compounding errors inherent in realistic recursive forecasting environments. Ultimately, the research underscores the profound autocorrelation within domestic electricity demand, demonstrating that uncomplicated baseline strategies can rival intricate algorithms in practical scenarios. These outcomes emphasize the critical role of rigorous evaluation methodologies and strategic feature engineering, offering empirical constraints on the deployment of advanced machine learning within real-world time-series applications.

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1 Introduction:

Predicting electricity utilization is a fundamental requirement for contemporary energy administration, facilitating load distribution, strategic demand planning, and economic optimization. As smart grid technologies expand, precise usage estimations are increasingly critical for both utility operators and end-users. However, projecting domestic power demand presents a formidable time-series challenge, given its inherent volatility driven by human behavioural fluctuations, daily habits, and exogenous variables (Hong et al., 2016). As illustrated in Figure 1, household electricity consumption exhibits clear daily patterns with peak demand occurring during morning and evening hours, reflecting typical human activity cycles.

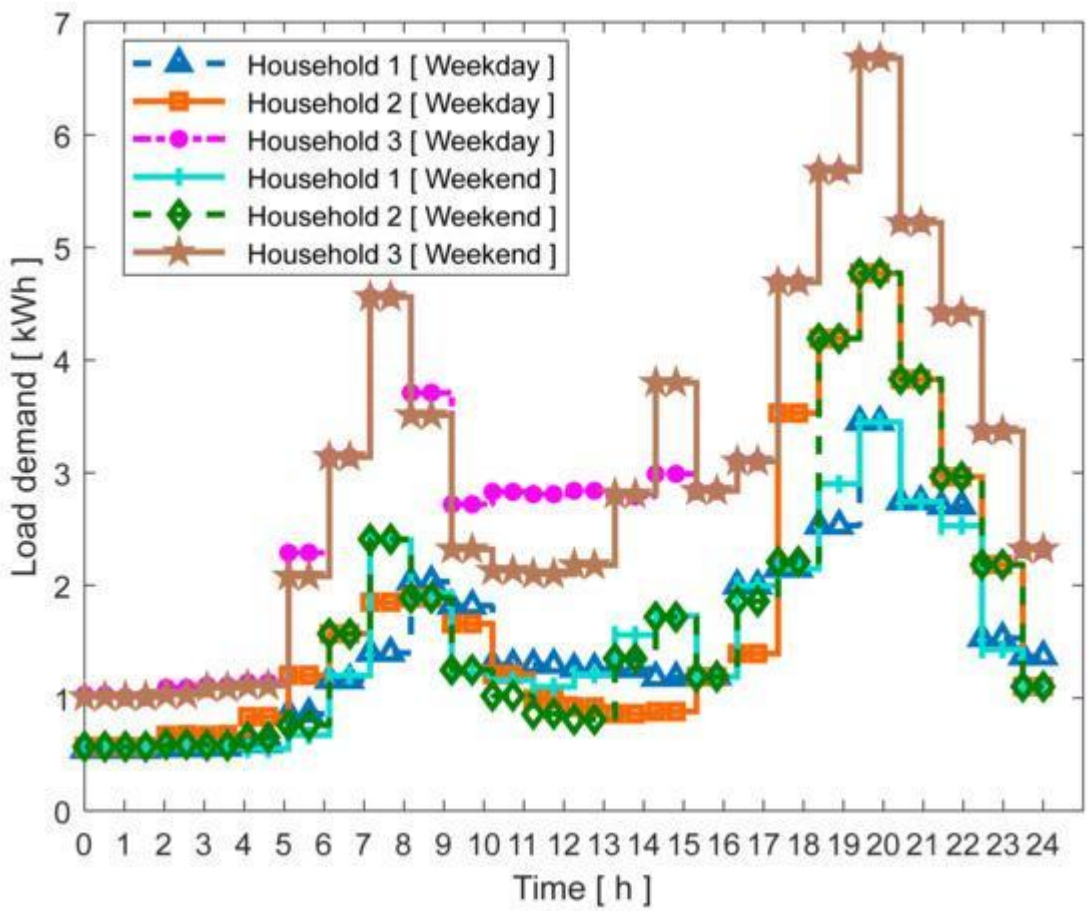


Figure 1: Hourly household electricity consumption patterns for weekdays and weekends, illustrating strong daily load variability and peak demand periods (source from MDPI Energies, Obaro et al., 2023).

Power consumption metrics are characterized by pronounced temporal correlations, manifesting as short-term inertia and cyclical patterns spanning hourly or daily horizons. Consequently, time-series methodologies are highly applicable to this domain. Traditional statistical architectures,

notably the Autoregressive Integrated Moving Average (ARIMA) and its seasonal variant (SARIMA), have historically been utilized for this purpose owing to their transparency and proficiency in mapping linear temporal dependencies (Box et al., 2015). Despite these strengths, classical frameworks frequently fail to accurately model the intricate, non-linear dynamics inherent in empirical consumption records. As shown in Figure 2, classical time-series models such as ARIMA and SARIMA can be extended with regression components to improve forecasting by incorporating additional explanatory variables.

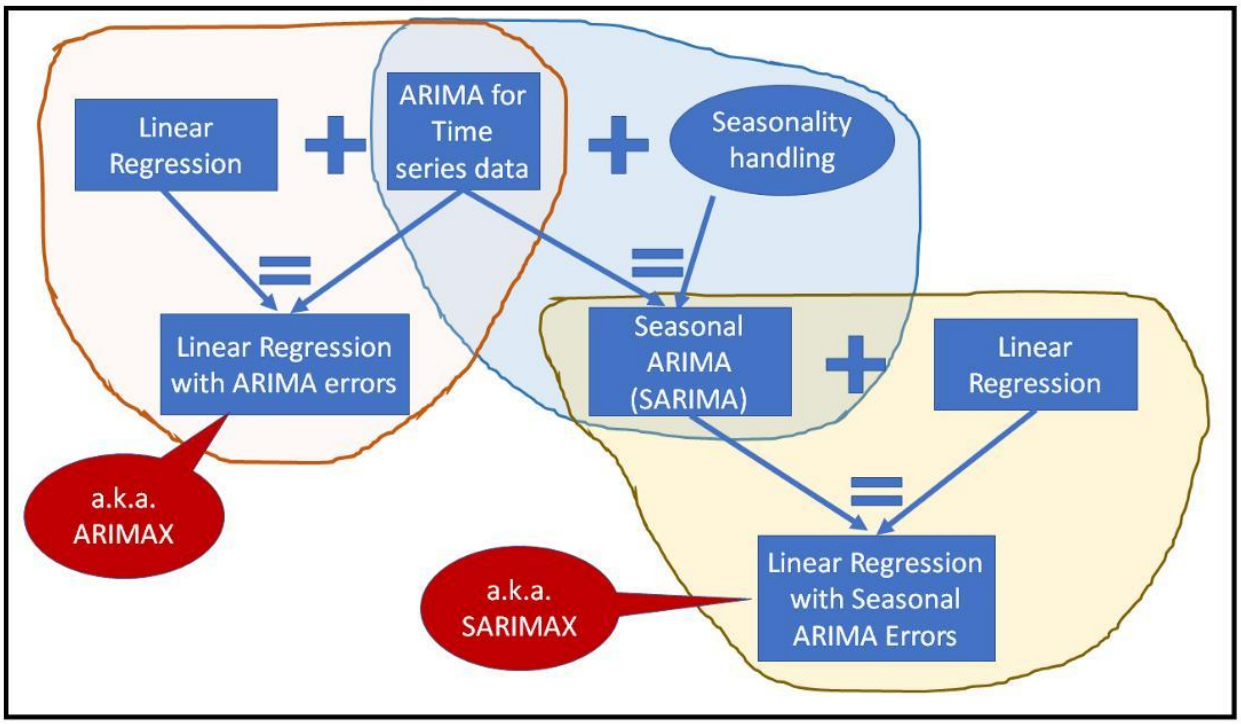


Figure 2: Conceptual framework of ARIMA, SARIMA, and their extensions (ARIMAX and SARIMAX), illustrating the integration of regression with time-series error modelling (source from Time Series Reasoning).

To address these limitations, machine learning paradigms have been increasingly adopted for predictive time-series analysis. Algorithms such as Random Forest and extreme gradient boosting (XGBoost) demonstrate superior efficacy in managing non-linearities and complex variable interactions. This is particularly evident when these models are augmented with engineered attributes, including rolling statistical windows and lag variables (Chen and Guestrin, 2016). Such algorithms consistently yield robust predictive accuracy when applied to historical, tabular time-series structures.

This research constructs a rigorous predictive framework targeting domestic energy use, utilizing the UCI Individual Household Electric Power Consumption dataset. The established analytical pipeline encompasses comprehensive data preprocessing, the resolution of omitted values via chronologically weighted interpolation, the aggregation of high-frequency telemetry into

standard hourly metrics, and the derivation of specialized temporal features. Utilizing this curated dataset, the study benchmarks traditional statistical frameworks against advanced machine learning algorithms.

A primary differentiator of this investigation is its rigorous evaluation paradigm. Beyond conventional single-step predictionswhich operate under the assumption of perfect historical data availabilitythis project implements a recursive forecasting protocol. In this iterative scenario, model outputs are systematically recycled as subsequent inputs. This approach yields a highly realistic appraisal of algorithmic stability in applied environments, specifically isolating how predictive deviations compound over extended time horizons.

This study is guided by the following research question:
Which time series modelling approaches provide the most accurate and robust forecasts of household electricity consumption when trained on realistically pre-processed data?

Ultimately, this research aims to determine the optimal algorithmic strategy for predicting residential power demand. Furthermore, it serves to empirically validate the critical influence of robust feature engineering, meticulous algorithm selection, and pragmatic evaluation protocols in generating dependable energy forecasts.Overall, this work seeks to identify the most suitable modelling approach for electricity consumption forecasting while highlighting the importance of feature engineering, model selection, and evaluation strategy in achieving reliable predictions.

2 Literature Review:

The optimization of smart grids, energy distribution, and demand-side planning relies heavily on accurate electricity consumption forecasting. The proliferation of smart metering infrastructure has transitioned research away from broad, aggregated datasets toward highly granular, household-level analysis (Hong et al., 2016). Consequently, contemporary literature frequently benchmarks both traditional statistical frameworks and advanced machine learning algorithms to effectively process this high-resolution telemetry.

Predictive accuracy is fundamentally constrained by the inherent complexities of household time-series data. Energy usage exhibits distinct cyclical patterns across hourly, daily, and weekly horizons, closely mirroring routine human behavior. Furthermore, temporal shifts in user habits and environmental variables cause the underlying statistical properties of the data to fluctuate continuously, leading to non-stationarity (Box et al., 2015). Unlike smoothed aggregate data, individual household telemetry is highly erratic, where the intermittent activation of high-draw appliances generates unpredictable consumption spikes. Additionally, hardware faults and transmission latency frequently result in missing values, necessitating rigorous imputation strategies to prevent algorithmic failure. Figure 2 demonstrates that electricity consumption is influenced by external factors such as temperature, indicating the importance of incorporating exogenous variables in forecasting models.

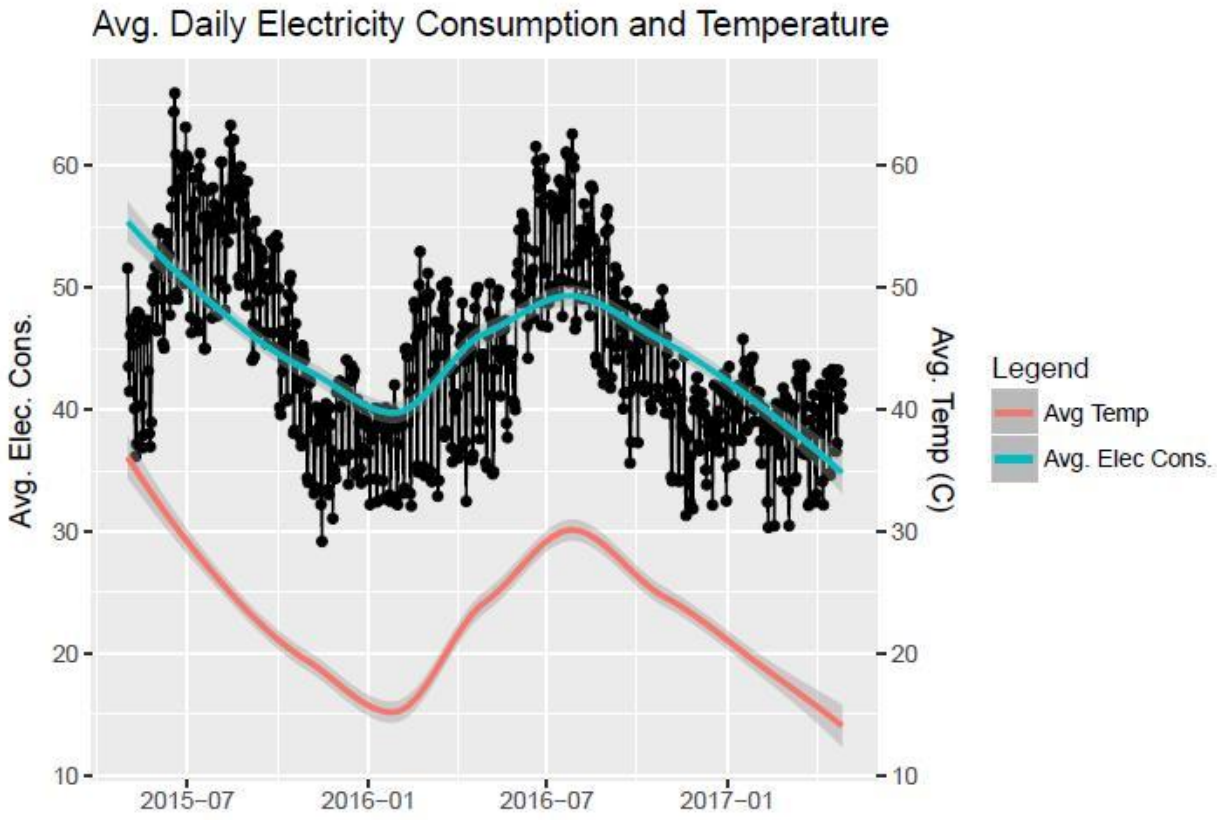


Figure 3: Relationship between average daily electricity consumption and temperature, highlighting seasonal trends and external environmental influence on energy usage (source from Medium, Koehrsen, 2018).

Empirical studies establish that diligent preprocessing is as critical as model selection for achieving high predictive fidelity. Time-weighted interpolation serves as the standard mechanism for resolving data gaps, preserving temporal continuity without injecting synthetic artifacts. Moreover, down-sampling high-frequency logs to broader temporal intervals is a proven technique for attenuating noise and optimizing computational efficiency. For machine learning architectures, historical context must be explicitly engineered through lag variables that capture immediate past dependencies, rolling metrics that quantify localized variance, and calendar parameters that map cyclical habits. Rigorous partitioning of training and testing data remains mandatory during these feature transformations to ensure robust statistical validity and prevent data leakage (Hyndman & Athanasopoulos, 2018).

In terms of methodological paradigms, classical architectures like ARIMA and SARIMA maintain a strong presence in forecasting literature due to their mathematical transparency and proven efficacy in mapping linear temporal relationships (Box et al., 2015). SARIMA, in particular, effectively accommodates recurrent cyclicality. However, these parametric models are heavily constrained by highly non-linear data and struggle to map the complex, multivariate interactions prevalent in volatile domestic usage profiles. To overcome these classical limitations, algorithms including Random Forest and extreme gradient boosting, specifically XGBoost, are increasingly utilized. These non-parametric models inherently process non-linear dynamics and complex feature interactions (Chen & Guestrin, 2016). While highly scalable, their efficacy is strictly dependent on the quality of engineered features, and they possess a higher susceptibility to overfitting without rigorous validation protocols. Figure 4 illustrates how machine learning models such as XGBoost utilize lagged observations as input features to capture temporal dependencies in time-series forecasting.

Figure 4: XGBoost-based time-series forecasting using lag features, demonstrating how past observations (t−1, t−7, t−30) are used to predict future values (source from Medium).

Despite the volume of existing literature, a critical methodological deficiency persists regarding the over-reliance on idealized evaluation metrics. The majority of studies benchmark algorithms utilizing single-step forecasting frameworks, which assume perpetual access to flawless historical data, thereby artificially inflating perceived model performance. Considerably less literature investigates realistic recursive forecasting scenarios, where iterative prediction errors accumulate over time. Furthermore, there is a notable scarcity of research that synthesizes data imputation, advanced feature engineering, and recursive evaluation into a single, reproducible pipeline tailored for individual households. Addressing this specific methodological gap constitutes the core motivation for this investigation.

3 Methodology:

This research adopts a clear, step-by-step data science framework to predict domestic electricity use. The design ensures the work can be easily reproduced, strictly prevents the leakage of future data into the training phase, and creates a fair testing environment to compare older statistical methods with modern machine learning algorithms. The complete process moves through data gathering, cleaning, variable creation, model building, and final testing.

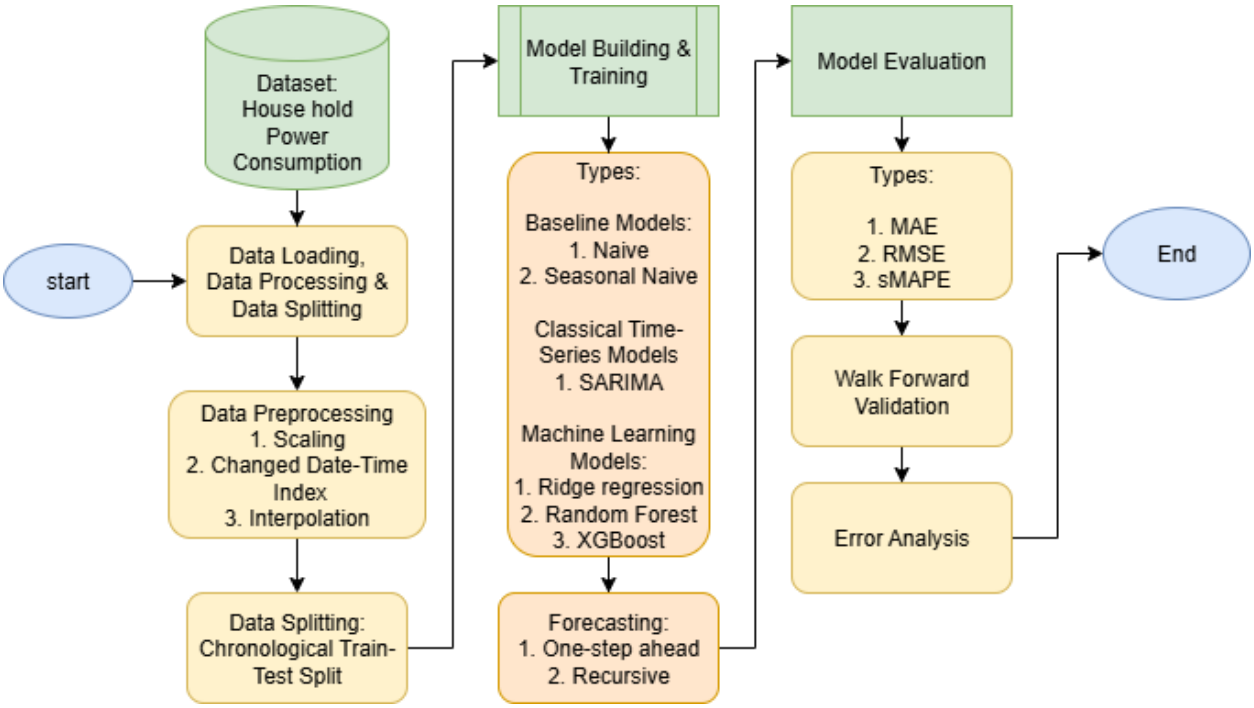
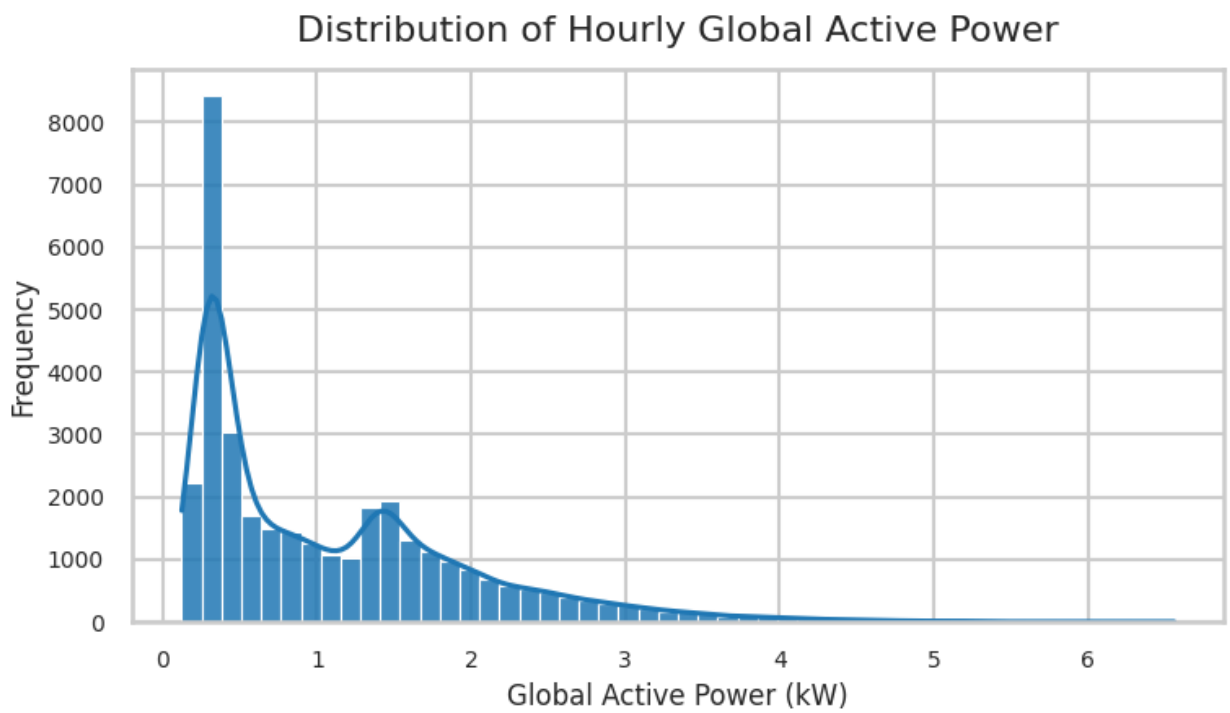


Figure 5: Complete Systsem Overview

3.1 Data Collection and Description

The analysis relies on the UCI Individual Household Electric Power Consumption dataset. This source provides minute-by-minute energy readings from a single home. The primary focus for prediction is "Global Active Power," which measures the total active electricity used in kilowatts.



With over two million individual data points, the dataset offers a very detailed look at how energy use changes over time. However, recording data every minute creates a lot of random noise and makes computer processing very slow. Therefore, the data must be cleaned and simplified before any models are built.

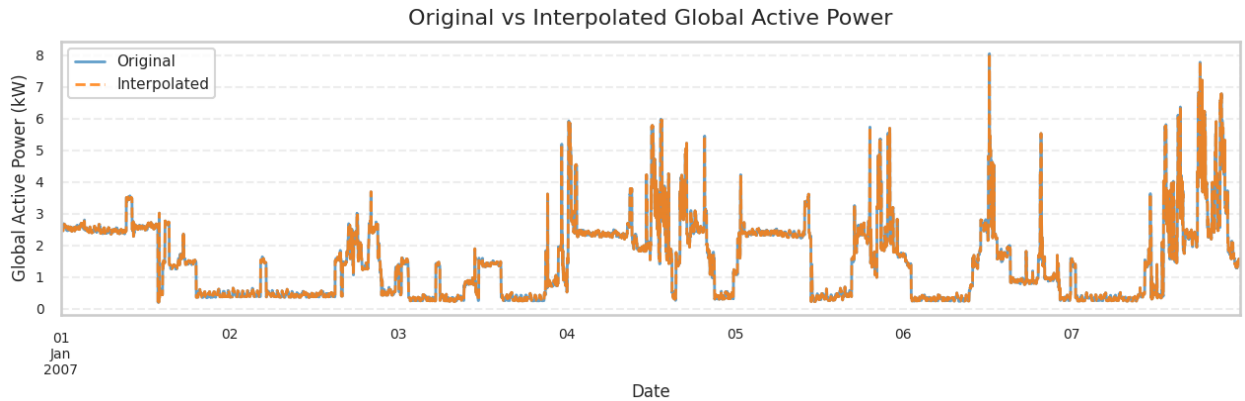
3.2 Data Preprocessing

3.2.1 Date-Time Transformation

Initially, the dataset records dates and times in separate columns. These are merged into a single, continuous timeline. This step is necessary to keep the data in the correct chronological order and to allow for time-based calculations later in the project.

3.2.2 Handling Missing Values

A small percentage of the data is missing. Because these gaps happen across multiple sensors at the same time, they are likely caused by brief equipment failures rather than permanent errors. To fix this, a time-based interpolation method is used to estimate the missing numbers based on the surrounding data, filling any remaining edge gaps by pulling nearby values forward or backward. This keeps the timeline smooth and avoids creating fake patterns.



3.2.3 Temporal Aggregation

To reduce the extreme fluctuations seen in minute-by-minute data, the readings are averaged into one-hour blocks. This step removes random noise, speeds up computer processing, and matches the standard approach for short-term energy forecasting while still keeping the most important daily patterns intact.

3.2.4 Feature Engineering

Creating new variables (features) from the existing data is a vital step, especially for machine learning models that need these inputs to understand time-based trends.

3.2.5 Calendar-Based Features

Specific time markerssuch as the hour of the day, the day of the week, the month, and whether it is a weekendare extracted from the timeline. These variables help the models recognize repeated human habits, like turning on lights every evening.

3.2.6 Lag Features

The models are given historical data points, called lags. By feeding the models the energy use from recent hours or the previous day, they can learn how past electricity use directly impacts immediate future use.

3.2.7 Rolling Statistics

To measure recent trends and shifts in behavior, rolling averages and standard deviations are calculated over specific time windows. It is critical that these calculations are shifted backward in time so the model only learns from the past, ensuring no future information accidentally influences the predictions.

3.2.8 Train-Test Splitting

The data is split into two sections using a strict chronological timeline. The first 80% of the timeline is used to train the models, and the final 20% is kept completely hidden to test them.

Keeping the data in chronological order is necessary to mimic a real-world scenario where the future is unknown.

3.3 Predictive Models

3.3.1 Baseline Models

Two basic models are created to set minimum performance standards. The "naive" model simply guesses that the next hour will be exactly the same as the current hour. The "seasonal naive" model guesses that the next hour will match the exact same hour from the day before. Complex models must beat these basic benchmarks to be considered useful.

3.3.2 Classical Time-Series Model

A Seasonal Autoregressive Integrated Moving Average (SARIMA) model is used to handle both linear trends and repeating daily cycles. After checking the data for statistical stability, this model is configured using standard mathematical practices and serves as the primary traditional benchmark.

3.3.3 Machine Learning Models

Advanced algorithms are used to find complex, non-linear patterns that traditional models might miss. Ridge Regression is applied to balance variables that might overlap too much. Random Forest is used to map out complicated relationships between different features, and XGBoost is included as a powerful, high-performance forecasting tool. Strict rules are followed to ensure these models only use past data when making predictions.

3.4 Forecasting Strategies

3.4.1 One-Step-Ahead Prediction

In this testing phase, the model predicts just one hour into the future while having perfect access to all actual past data. This represents an ideal, controlled environment to see the maximum potential of the algorithm.

3.4.2 Recursive Forecasting

To simulate a real-world situation where true future data is hidden, the models must make continuous predictions. They do this by feeding their own guesses back into themselves to predict the next steps. This is a much harder test because small mistakes tend to add up and grow over time.

3.5 Model Evaluation

3.5.1 Evaluation Metrics

The accuracy of each model is judged using standard mathematical scores: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Symmetric Mean Absolute Percentage Error (sMAPE). Together, these scores measure general accuracy, penalize large mistakes, and show proportional errors.

3.5.2 Walk-Forward Validation

To make sure the models remain reliable as time passes, they are tested using a moving window. The model trains on a set amount of data, tests on the next block, and then moves forward to repeat the process. This proves the model is stable and not just memorizing one specific timeframe.

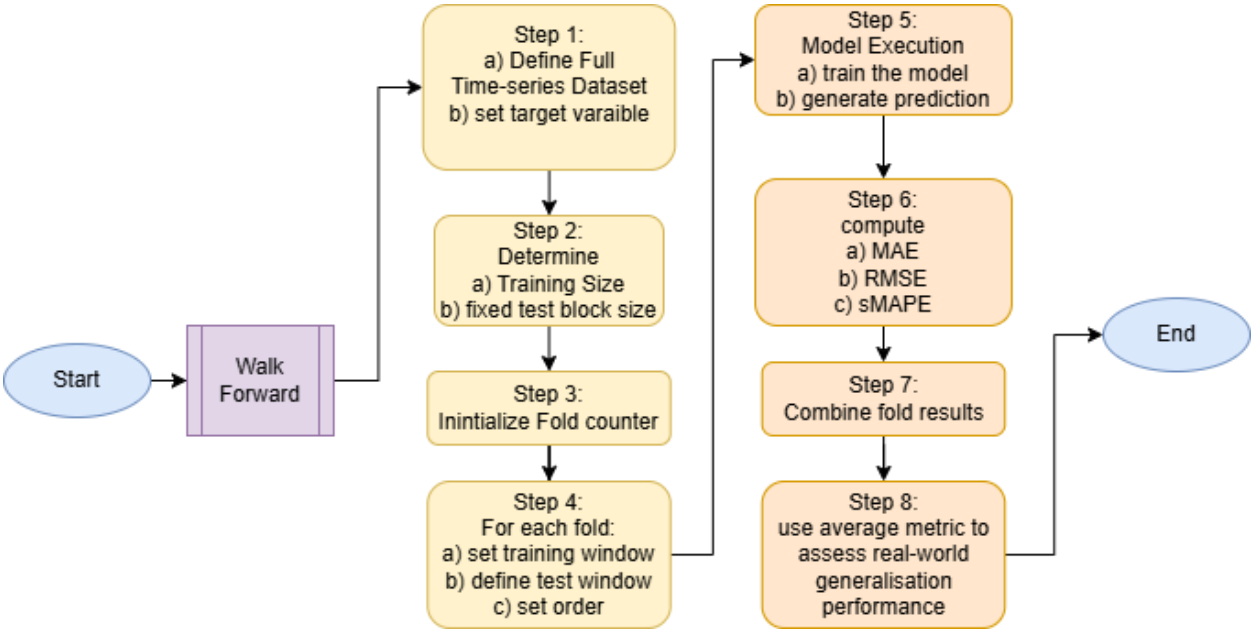


Figure 6: Complete Wak Forward System Overview

3.5.3 Error Analysis

Finally, the mistakes made by the models are analyzed closely. By looking at where and how the models failed, it becomes clear whether they actually learned the core habits of the household or if they were just distracted by random daily noise.

4 Results

This section presents the performance of the forecasting models developed in this study under multiple evaluation settings. The results are structured to compare baseline models, classical time-series approaches, and machine learning models using one-step-ahead prediction, recursive forecasting, and walk-forward validation. The objective is to provide a comprehensive and fair assessment of model performance under both ideal and realistic conditions

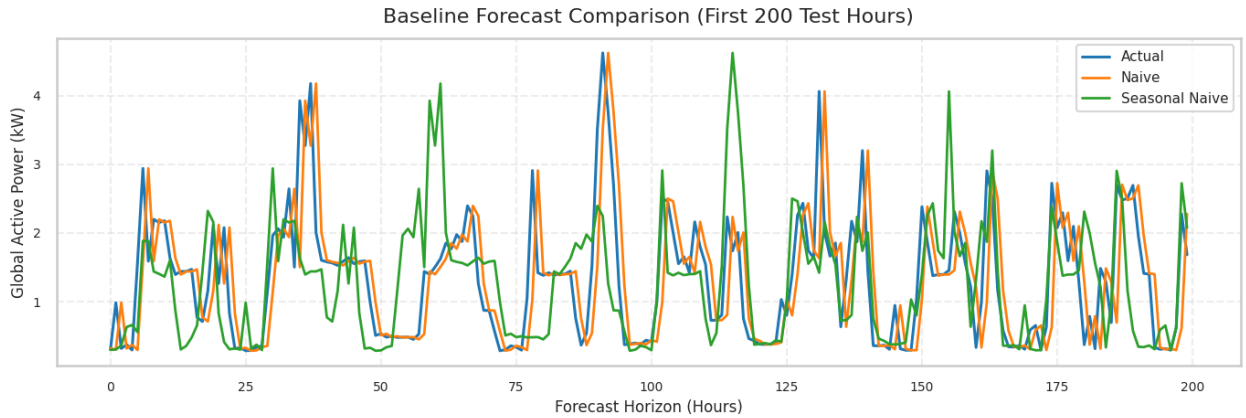


Figure 7: Baseline Forecast Comparison (First 200 Test Hours) - Naive and Seasonal Naive

4.1 One-Step-Ahead Model Performance

The one-step-ahead evaluation represents an ideal forecasting scenario where true historical values are available at each prediction step. Under this setting, machine learning models demonstrate superior performance compared to baseline and classical approaches.

Table 1: One Step Ahead Model Performance

Model	MAE	RMSE	sMAPE (%)
XGBoost	0.322574	0.462059	33.732713
RandomForest	0.335920	0.478809	34.265371
Ridge	0.363217	0.514455	37.934526
Naive	0.381306	0.589806	36.812807
SARIMA	0.514522	0.683324	51.333524
Seasonal Naive (24)	0.525511	0.778032	49.165095

The results indicate that XGBoost achieves the lowest error across all metrics, followed by Random Forest and Ridge Regression. Baseline models, particularly the naïve approach, remain competitive, outperforming SARIMA in terms of error metrics. The seasonal naïve model performs the worst, suggesting that daily seasonality alone is insufficient to capture the underlying consumption dynamics.

4.2 Recursive Forecasting Results

Recursive forecasting provides a more realistic evaluation by feeding predicted values back into the model. This introduces error accumulation, making the task significantly more challenging.

Table 2: Recursive Forecasting Results

Model	MAE	RMSE	sMAPE (%)
XGBoost (recursive)	0.519145	0.689998	50.942086
Random Forest (recursive)	0.574133	0.707237	58.426289

The results show a substantial degradation in performance for both models compared to the one-step-ahead setting. XGBoost remains the better-performing model, but its advantage over classical methods is reduced. This highlights the impact of error propagation in multi-step forecasting scenarios.

4.3 Walk-Forward Validation Results (Naive)

Walk-forward validation evaluates model stability across different time segments using an expanding training window.

Table 3: Walk Forward Validation Results (Naive)

Fold	Model	MAE	RMSE	sMAPE
1	Naive	0.341509	0.545613	37.953280
2	Naive	0.480516	0.709505	37.844104
3	Naive	0.401899	0.614024	36.878581

The naïve model demonstrates consistent performance across folds, indicating strong stability and robustness over time.

Table 4: Walk Forward Validation Results (SARIMA)

Fold	Model	MAE	RMSE	sMAPE
1	SARIMA	0.438243	0.579112	52.803069
2	SARIMA	0.704083	0.998038	58.510837
3	SARIMA	0.508027	0.674905	47.445159

SARIMA exhibits higher variability across folds, particularly in the second fold, where error metrics increase significantly. This suggests sensitivity to changing data patterns.

4.4 Walk-Forward Validation Results (Recursive XGBoost)

Table 5: Walk Forward Validation Results (Recursive XGBoost)

Fold	Model	MAE	RMSE	sMAPE
1	Recursive XGBoost	0.511733	0.673847	56.693046
2	Recursive XGBoost	0.552680	0.760714	44.895741
3	Recursive XGBoost	0.493872	0.660592	46.958885

Recursive XGBoost demonstrates moderate stability across folds, with improved performance in later segments, indicating better adaptation to temporal patterns.

4.5 Combined Walk-Forward Results

Table 6: Combined Walk-Forward Results

Fold	Model	MAE	RMSE	sMAPE
1	Naive	0.341509	0.545613	37.953280
2	Naive	0.480516	0.709505	37.844104
3	Naive	0.401899	0.614024	36.878581
1	SARIMA	0.438243	0.579112	52.803069
2	SARIMA	0.704083	0.998038	58.510837
3	SARIMA	0.508027	0.674905	47.445159
1	Recursive XGBoost	0.511733	0.673847	56.693046
2	Recursive XGBoost	0.552680	0.760714	44.895741
3	Recursive XGBoost	0.493872	0.660592	46.958885

This combined view highlights the comparative behaviour of models across time, reinforcing the stability of the naïve model and the variability of SARIMA and recursive machine learning approaches.

4.6 Average Walk-Forward Performance

Table 7: Average Walk-Forward Performance

Model	MAE	RMSE	sMAPE
Naive	0.407975	0.623047	37.558655
Recursive XGBoost	0.519429	0.698384	49.515891
SARIMA	0.550118	0.750685	52.919688

On average, the naïve model outperforms both SARIMA and recursive XGBoost, confirming its effectiveness as a strong baseline.

4.7 Overall Model Comparison

Table 8: Overall Model Comparison

Model	MAE	RMSE	sMAPE	Evaluation Type
XGBoost (1-step)	0.322574	0.462059	33.732713	1-step ML
Random Forest (1-step)	0.335920	0.478809	34.265371	1-step ML
Ridge (1-step)	0.363217	0.514455	37.934526	1-step ML
Naive	0.381306	0.589806	36.812807	Baseline
SARIMA	0.514522	0.683324	51.333524	Classical
XGBoost (recursive)	0.519145	0.689998	50.942086	Recursive ML
Random Forest (recursive)	0.574133	0.707237	58.426289	Recursive ML
Seasonal Naive	0.525511	0.778032	49.165095	Baseline

The comparison clearly shows that one-step machine learning models outperform all other approaches. However, under realistic recursive conditions, their performance becomes comparable to classical models.

4.8 Best Model by Metric

- Model XGBoost (1-step)
- MAE 0.322574
- RMSE 0.462059
- sMAPE 33.732713
- Evaluation_Type 1-step ML

XGBoost consistently achieves the best performance across all evaluation metrics under the one-step-ahead setting.

4.9 Visual Comparison of Model Performance

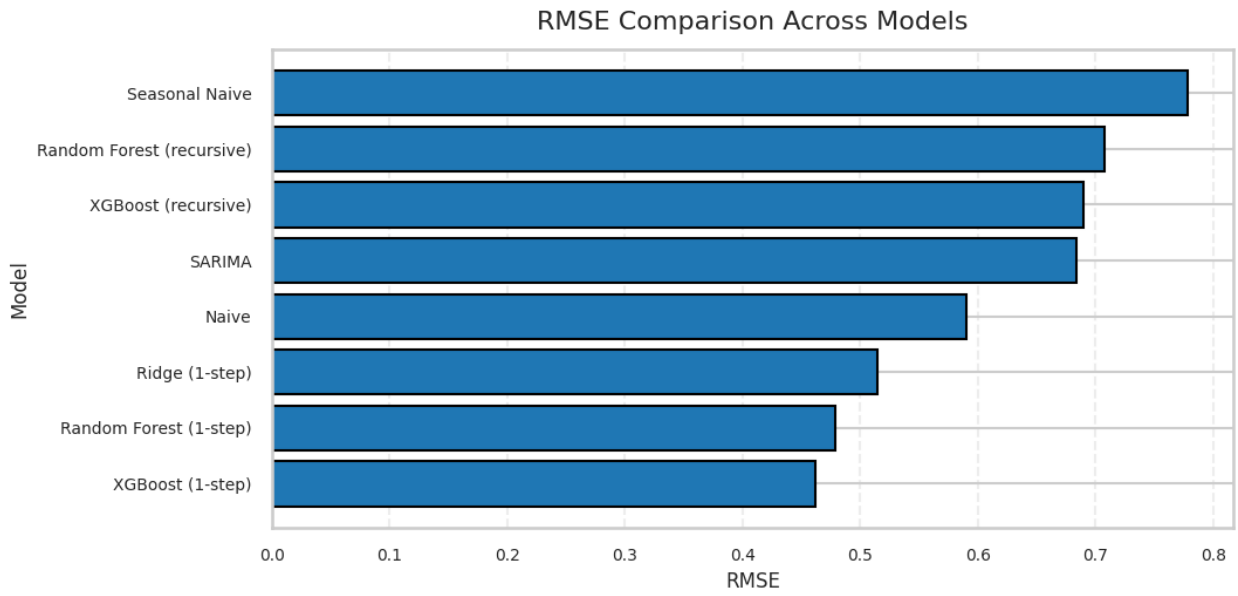


Figure 8: RMSE Comparison Across Models

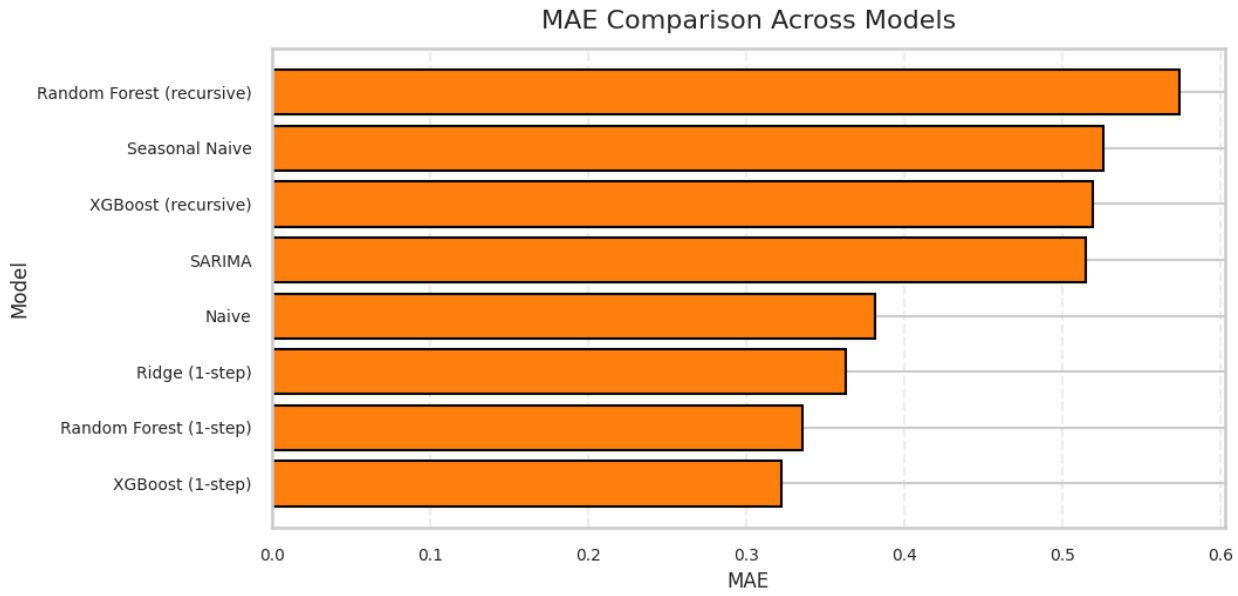


Figure 9: MAE Comparison Across Models

5 Conclusion

‘This research conducted a thorough investigation into the prediction of domestic energy demand, contrasting traditional statistical techniques with contemporary machine learning algorithms within a rigorous and pragmatic testing environment. The project successfully engineered a comprehensive analytical pipeline, transitioning from raw high-frequency telemetry through data cleaning, feature extraction, algorithmic training, and multi-layered testing. The empirical findings establish that residential power usage is heavily dictated by immediate historical precedent, a reality underscored by the surprisingly robust performance of the rudimentary naive baseline. When evaluated under idealized single-step conditions, advanced algorithms, specifically XGBoost, demonstrated superior predictive precision across all mathematical metrics. However, when subjected to the iterative recursive testing framework, which accurately simulates actual operational deployment, the efficacy of these sophisticated models degraded notably due to compounding predictive errors. In this practical scenario, the performance disparity between complex machine learning frameworks, conventional statistical models, and basic baselines narrowed significantly. Furthermore, walk-forward validation confirmed these observations, illustrating that the naive benchmark retained reliable stability across varying temporal windows, whereas SARIMA and recursive machine learning outputs demonstrated pronounced volatility. Ultimately, this investigation proves that algorithmic complexity does not guarantee proportional gains in forecasting precision, particularly within datasets characterized by extreme temporal autocorrelation.

5.1 Limitations

Despite the comprehensive nature of this analysis, several methodological constraints must be acknowledged. The empirical data was sourced exclusively from a solitary residence, fundamentally restricting the capacity to generalize these findings across wider demographic populations or diverse geographic regions. Furthermore, the analysis excluded critical exogenous variables, such as meteorological conditions, household occupancy dynamics, and specific appliance activation cycles, thereby preventing the models from mapping external drivers of energy demand. Methodologically, the machine learning algorithms depended entirely on manually derived temporal features, which may fail to encapsulate the absolute complexity of underlying time-series relationships. The research was also restricted to short-term predictive horizons, primarily because the adopted recursive forecasting methodology inherently propagates errors, rendering long-term projections unreliable. Finally, due to computational resource limitations, exhaustive hyperparameter optimization was not conducted, suggesting that the absolute maximum performance threshold of certain algorithms may remain unrealized.

5.2 Future Scope

Subsequent research initiatives can resolve these constraints by integrating diverse external data streams, particularly meteorological records, real-time occupancy metrics, and socioeconomic indicators, thereby enriching the predictive context. Broadening the empirical foundation to encompass aggregated data from numerous households or entire geographic districts would significantly improve the generalizability of the models and facilitate the design of resilient, scalable forecasting architectures. Regarding algorithmic advancement, future studies should investigate deep learning paradigms, notably Long Short-Term Memory networks and Transformer architectures, which possess the intrinsic capacity to extract complex temporal dynamics without relying on exhaustive manual feature derivation. The development of hybrid

frameworks, synthesizing the mathematical stability of traditional time-series methods with the non-linear capabilities of machine learning, also presents a promising avenue for performance enhancement. Additional methodological improvements should include advanced hyperparameter optimization protocols and the application of probabilistic forecasting techniques to formally quantify predictive uncertainty. Moreover, replacing the iterative recursive evaluation strategy with direct multi-step forecasting could effectively eliminate the compounding error propagation observed in this study. In summary, while this research confirms that sophisticated algorithms achieve exceptional accuracy under theoretical conditions, it simultaneously proves that practical deployment necessitates rigorous evaluation frameworks and respects the persistent viability of fundamental baseline methodologies.

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7 Appendix

Appendix A:

Below is the complete source code of the project.

[Project Code]

Appendix B:

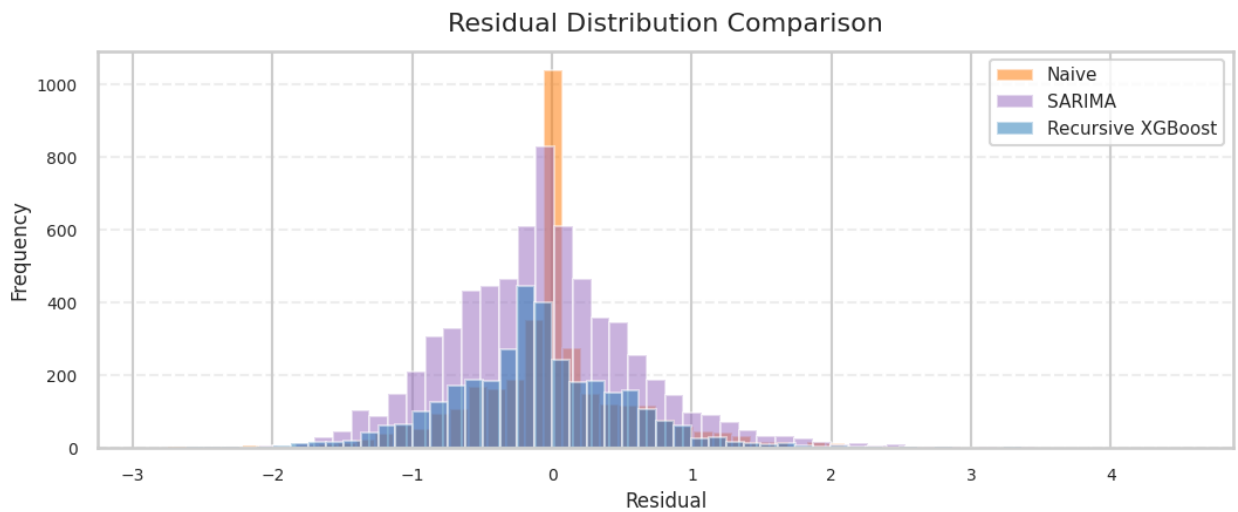

Figure: Residual Distribution Comparision

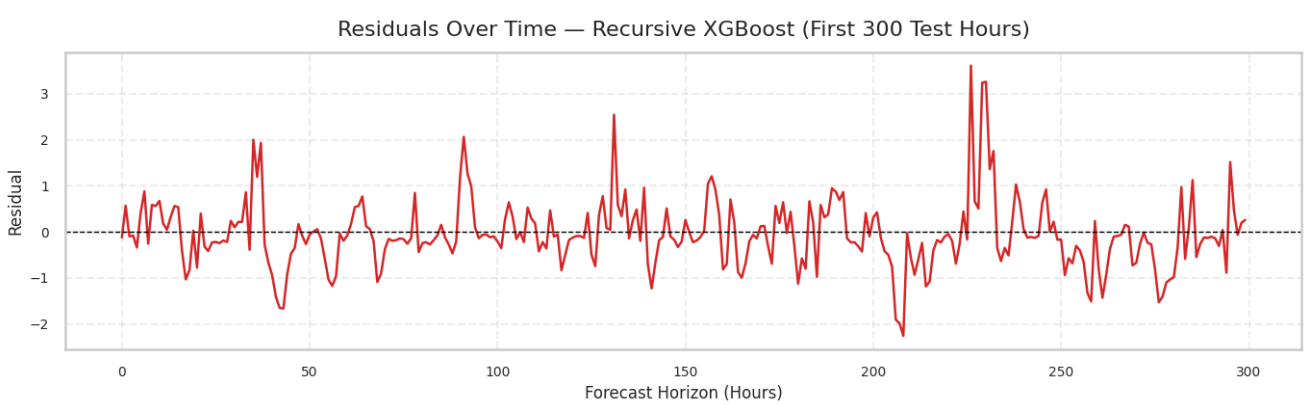

Figure: Residual Over Time – Recursive XGBoost

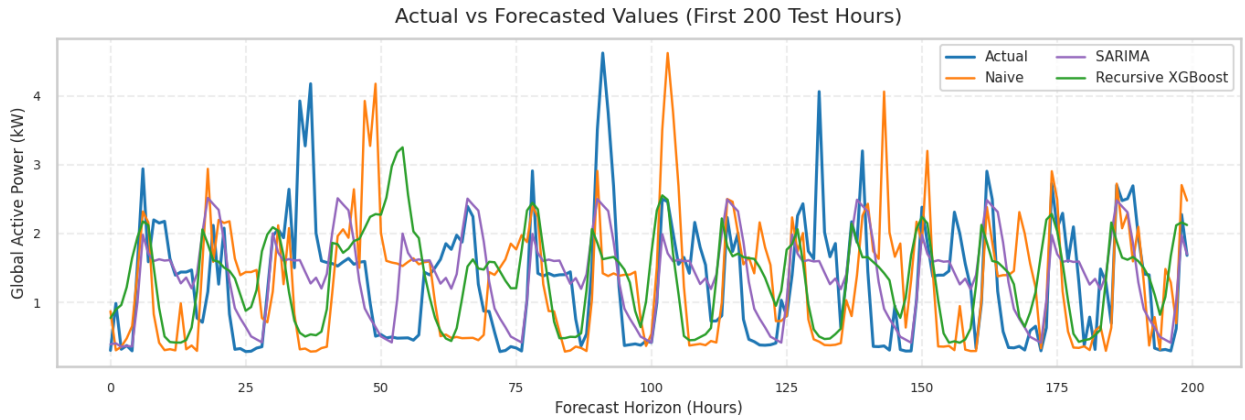

Figure: Actual vs Predicted Values.

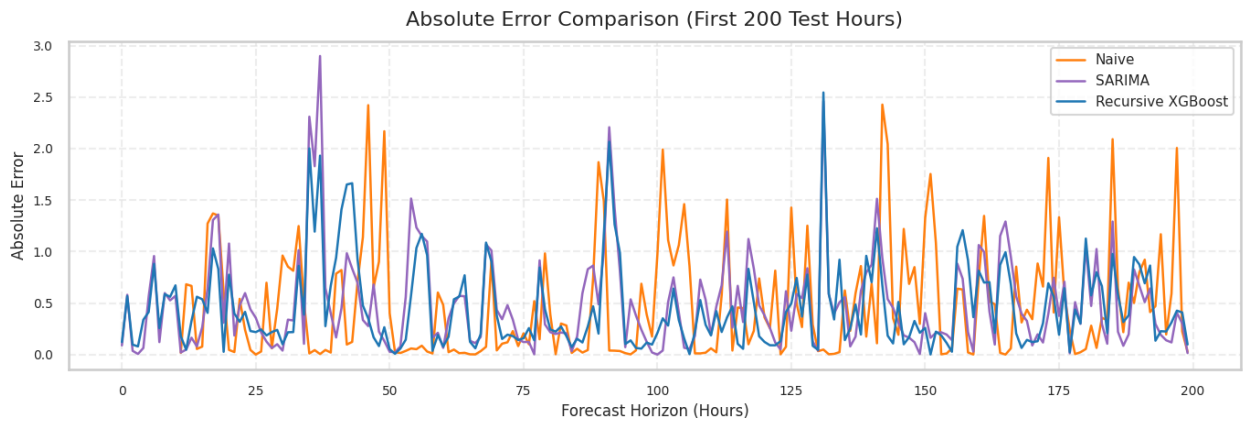

Figure: Absolute Error Comparision.